

3. Malaria kills a large number of people in the world each year. It is caused by the single celled organism called *Plasmodium*, which is spread by female mosquitoes (*Anopheles gambiae*).



Female mosquito (*Anopheles gambiae*)

In 1898, some Italian scientists developed a method to control malaria which they called bonifica. They used this approach in a small test area in Italy.

Bonifica involved:

- spraying insecticides in people’s houses;
- hanging mosquito nets in windows of houses;
- pouring petrol on the water in the marshes, where mosquitoes reproduce.

There were 99 new cases of malaria in the test area during the year before bonifica. In the year after there were no new cases.

More recently, anti-malarial tablets have been developed along with a vaccine against *Plasmodium* and the use of biological control.

- (a) (i) Use the information above to complete the following table by writing True or False against **each** statement. [3]

Statement	True or False
In 1899 there were no new malaria patients in the test area.	
Insecticides kill <i>Plasmodium</i> .	
There is evidence that bonifica was successful.	
<i>Plasmodium</i> is passed directly from person to person.	

- (ii) State the genus of the mosquito that spreads *Plasmodium*. [1]

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(b) (i) State the purpose of the nets that were put up at the window. [1]

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(ii) Suggest why bonifica could be dangerous to the people who live near the marshes where the mosquitoes reproduce. [1]

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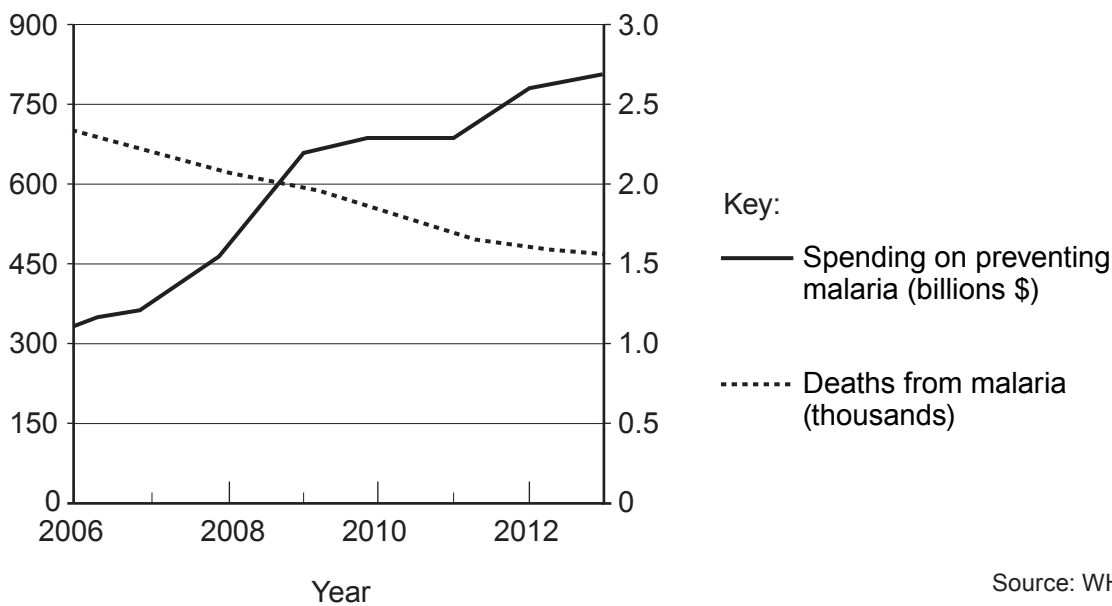
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(c) **Graph 3.1** shows the number of annual global deaths from malaria and the amount of money spent on preventing malaria from 2006 to 2013.

Graph 3.1

Deaths from malaria
(thousands)

Spending on preventing
malaria (billions \$)



State the relationship between the money spent on preventing malaria and the number of deaths from malaria. [1]

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- (d) Some of the modern methods used to control malaria include the use of vaccines and biological control.
Choose words from the list to complete the following sentences. [2]

starving preventing killing protecting

Vaccines work by a person getting malaria.
Biological control works by the mosquito that is spreading malaria.

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